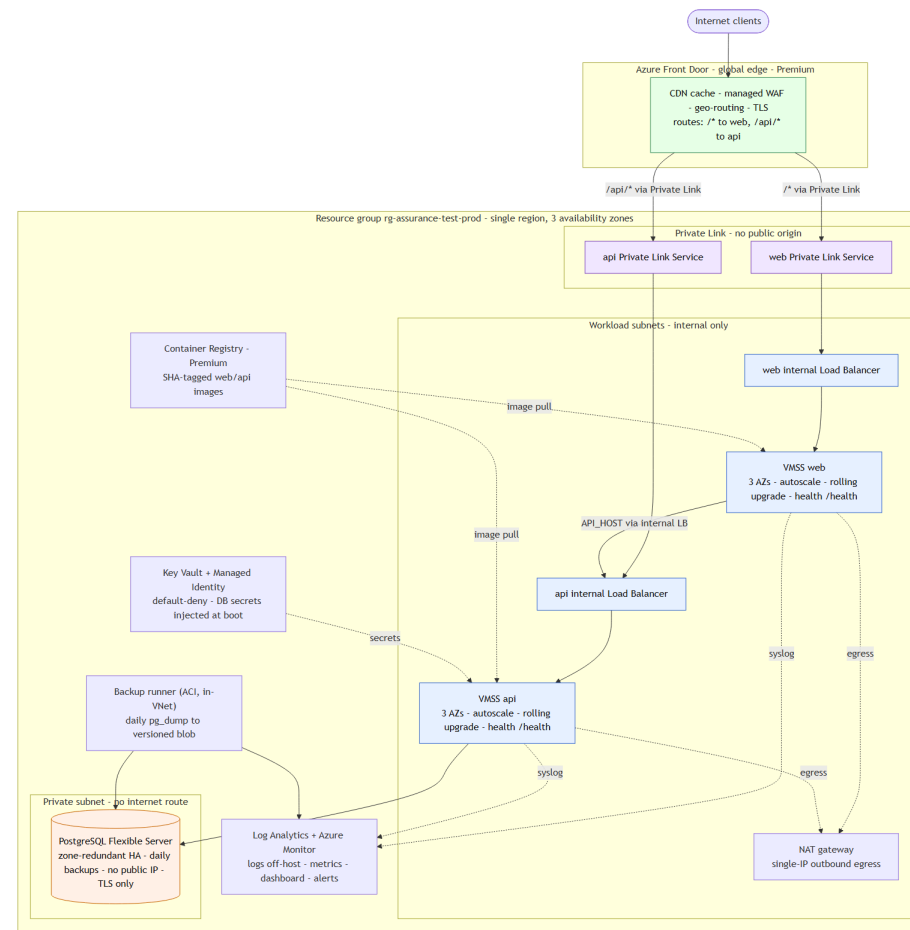


Continuous delivery for a 3-tier Node platform

Scalable, secure, self-healing, zero-downtime. On Azure, fully IaC.

Architecture



Azure Front Door reaches VMSS web and api privately over Private Link across three zones, to a private zone-redundant PostgreSQL.

How each requirement is met

Requirement	Implementation
Web and API on the Internet	Front Door routes <code>/*</code> and <code>/api/*</code> to per-tier VMSS behind internal load balancers, reached privately over Private Link
Origins never public	Internal load balancers exposed to Front Door only through a Private Link Service; instances have no public IP; a NAT gateway provides outbound egress
Database not public	PostgreSQL Flexible Server, private VNet access, NSG-restricted, no public IP
Fully IaC	Terraform, remote state with locking, one resource group, three environments
Survive instance failure	VMSS across two zones, health probes, auto-repair; zone-redundant database HA
Update without downtime	Health-gated VMSS rolling upgrades, SHA-tagged images
Automated deployment	GitHub Actions: test, build, apply, rolling upgrade, smoke test
Daily backups	Flexible Server point-in-time restore plus an in-VNet pg_dump to versioned blob
Logs off-host	Azure Monitor Agent and diagnostics to Log Analytics
Historical metrics	Azure Monitor dashboard and alerts
CDN by location	Azure Front Door global edge with static caching

Delivery pipeline and resilience

- **Pipeline:** push, then lint and test, then build and push SHA-tagged images to ACR, then terraform apply, then a health-gated rolling upgrade, then a smoke test through Front Door.
- **Zero downtime:** a new image replaces an instance only after the new one passes `/health`; Front Door and the load balancer route only to healthy nodes.
- **Self-healing:** unhealthy instances are auto-repaired; the database fails over to a standby in another zone.
- **Secrets:** Key Vault plus managed identity, no credentials on disk or in git.
- **State:** shared Terraform state in Azure Storage with blob-lease locking, ready for a team.

Security posture

- **Edge:** Premium Front Door with the managed OWASP ruleset and bot-manager rules in blocking mode.
- **Private origins:** the tiers sit behind internal load balancers exposed to Front Door only through a Private Link Service, so no origin is reachable from the public internet; a NAT gateway gives the instances controlled egress.
- **Data planes locked:** Key Vault, the backup storage account and the Terraform state account default-deny public access and are reachable only from the app subnets and the CI runner.
- **Secrets and transport:** secrets come from Key Vault via managed identity, never from disk or git; the database connection is TLS-only.
- **Supply chain:** keyless OIDC deploys, workflow actions pinned to commit digests, and the image scan runs in a job with no cloud credentials.
- **Access:** Entra ID role groups mapped to least-privilege scoped roles; the database administrator is a group, not a shared password.

Observability

- **Central logs:** the Azure Monitor Agent ships host and container syslog off-box to Log Analytics.
- **Health:** a synthetic web test hits `/health` from multiple regions and raises an availability alert on failure.
- **Alerts:** metric alerts on web, api and database CPU, on Front Door 5xx and latency, and on database storage, all wired to an email action group.
- **Backups watched:** a scheduled-query alert fires if a fresh database dump does not land in the blob.
- **Cost control:** workspace ingestion is capped per environment with 30-day retention.

One app, three deployment models

- **VMSS (reference):** VM Scale Sets running containers, the deployed and verified primary.
- **AKS and Helm:** the same app on Kubernetes, with Deployments, HPA, rolling updates and zone spread.
- **Azure Container Apps:** serverless containers with built-in ingress, revisions and scale-to-zero.

VMSS is the primary because it maps directly to starting, stopping and scaling nodes and keeps the moving parts small. The alternatives show the same architecture on other platforms.